



UNIVERSITY OF SRI JAYEWARDENEPURA - FACULTY OF APPLIED SCIENCES
BSc Degree Second Year First Semester Course Unit Examination – October/November 2022

DEPARTMENT OF MATHEMATICS

MAT 211 1.0 Linear Algebra II

Time: One (01) hour

No. of questions: 05

No. of pages: 02

Total marks: 80

Instructions:

- Answer all questions.
- This paper carries a total of 80 marks.
- You may bring **one A4 Sheet** of information for the exam.
- Paper has three sections.

Section A. You need to indicate whether the statement is True or False. Justification is not necessary (**10 marks**). Each question carries **two (2) marks** in the following way:

2 marks for a correct answer

-2 for a wrong answer

0 for not answering (when you do not know the answer).

Section B. State whether the statement is True or False, and then justify your answer. (**15 marks**).

Section C has 3 short essay type questions (**55 marks**).

- Usual notations apply throughout the paper.
 E_λ denotes the eigen space of the eigenvalue λ .

Section A

1. State whether the following statements are true or false. Justification is not necessary.

(i) Let $K = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Then $K^5 = K$.

(ii) Let A and B be square matrices of the same order. If $\det A = \det B$, then $A = B$.

(iii) $A = \begin{pmatrix} -2 & 5 & -1 \\ 0 & 4 & 2 \\ 0 & 0 & -1 \end{pmatrix}$ is a diagonalizable matrix.

(iv) Let $A = \begin{pmatrix} 4 & 3 & 7 \\ 0 & -2 & 4 \\ 0 & 0 & 3 \end{pmatrix}$. Then $\det(A + 2I) = 0$.

(v) Take the matrix $\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$. Then $\dim E_4 = 1$.

[10]

Section B

2. Prove or disprove the following statements.

- (i) A diagonalizable matrix A has characteristic equation $(\lambda - 4)^3 = 0$, then $A = 4I$. [5]
- (ii) There exists a matrix B with an eigenvalue λ with $\dim E_\lambda = 0$. [5]
- (iii) If two matrices are similar, they have the same eigenvalues. [5]

Section C

3. (i) What is meant by a square matrix is

- (a) diagonalisable; [3]
- (b) orthogonally diagonalizable. [3]

(ii) (a) Express the quadratic expression $f(x, y) = 3x^2 + 2xy + 3y^2$ in the form $X^T A X$, where A is a symmetric matrix. [6]

(b) Give reasons as why the matrix A is orthogonally diagonalizable. [5]

(c) Hence, identify the conic section given by $3x^2 + 2xy + 3y^2 = 1$. [18]

4. Suppose the matrix A has the characteristic polynomial

$$p(\lambda) = (\lambda - 1)^2(\lambda - 4)^2(\lambda - 8)$$

with geometric multiplicity of $\lambda = 1$ is 2, and geometric multiplicity of $\lambda = 4$ is 1.

Without doing any calculation, determine a possible Jordan canonical form of A .

[9]

5. Use Gram-Schmidt process to obtain an orthonormal basis from the basis $\{(1, 2), (2, 1)\}$ of $\mathbb{R}^2$.

[11]

